

Deep Jump Gaussian Processes for Surrogate Modeling of High-Dimensional Piecewise Continuous Functions

Appendix A. Derivation of Closed Form of ELBO

In this appendix, we provide the full derivation of the evidence lower bound (ELBO) for the DJGP model, using the notation and variational family adopted in the main text. Throughout, $j \in \{1, \dots, J\}$ indexes test regions, $i \in \mathcal{D}_n^{(j)}$ indexes local neighbors, $\ell \in \{1, \dots, L_1\}$ (local inducing) and $\ell \in \{1, \dots, L_2\}$ (global inducing), $k \in \{1, \dots, K\}$ indexes latent coordinates, and $d \in \{1, \dots, D\}$ indexes observed dimensions.

Variational factorization. We approximate the posterior by

$$q(\{\mathbf{f}^{(j)}, \mathbf{r}^{(j)}, \mathbf{v}^{(j)}, \mathbf{W}_j\}_{j=1}^J, \mathbf{R}) = \prod_{j=1}^J \left[p(\mathbf{f}^{(j)} \mid \mathbf{r}^{(j)}, \mathbf{W}_j, \boldsymbol{\Theta}^{(j)}) q(\mathbf{r}^{(j)}) \prod_{i \in \mathcal{D}_n^{(j)}} q(v_i^{(j)}) \right] p(\mathbf{W} \mid \mathbf{R}, \boldsymbol{\Theta}_W) q(\mathbf{R}),$$

with

$$q(\mathbf{r}^{(j)}) = \mathcal{N}(\boldsymbol{\mu}_r^{(j)}, \boldsymbol{\Sigma}_r^{(j)}), \quad q(v_i^{(j)}) = \text{Bernoulli}(\rho_i^{(j)}),$$

and the **mean-field per-element** global inducing posterior

$$q(\mathbf{R}) = \prod_{\ell=1}^{L_2} \prod_{k=1}^K \prod_{d=1}^D \mathcal{N}(R_{\ell,k,d} \mid \mu_{\ell kd}, \sigma_{\ell kd}^2). \quad (1)$$

Variational distribution of $\mathbf{W}$. We do not introduce an explicit variational factor for the projection matrices $\mathbf{W}$. Instead, $\mathbf{W}$ follows the conditional GP prior $p(\mathbf{W} \mid \mathbf{R}, \boldsymbol{\Theta}_W)$ under the global variational posterior $q(\mathbf{R})$:

$$q(\mathbf{W}) = \mathbb{E}_{q(\mathbf{R})}[p(\mathbf{W} \mid \mathbf{R}, \boldsymbol{\Theta}_W)] = \prod_{j=1}^J \mathbb{E}_{q(\mathbf{R})}[p(\mathbf{W}_j \mid \mathbf{R}, \boldsymbol{\Theta}_W)]. \quad (2)$$

Given $\mathbf{R}$, the local projections $\{\mathbf{W}_j\}_{j=1}^J$ are conditionally independent. Because both $p(\mathbf{W} | \mathbf{R}, \Theta_W)$ and $q(\mathbf{R})$ are Gaussian, the induced marginal $q(\mathbf{W})$ and each $q(\mathbf{W}_j)$ are also Gaussian.

Induced marginal $q(\mathbf{W}_j)$ and its moments. Under the global GP prior

$$p(\mathbf{W} | \Theta_W) = \prod_{k=1}^K \prod_{d=1}^D \mathcal{N}(\mathbf{w}_{kd} | \mathbf{0}_J, \mathbf{C}_w^{(k)}), \quad (3)$$

each coordinate process $\mathbf{w}_{kd} = [w_{kd}^{(1)}, \dots, w_{kd}^{(J)}]^\top$ is a zero-mean Gaussian process with covariance $[\mathbf{C}_w^{(k)}]_{jj'} = s^2 \exp(-\frac{1}{2} \|\mathbf{x}_*^{(j)} - \mathbf{x}_*^{(j')}\|^2 / \ell_{w,k}^2)$, where $\Theta_W = (s, \ell_{w,1}, \dots, \ell_{w,K})$. Let $\mathbf{R}_{:kd} = [R_{1kd}, \dots, R_{L_2 kd}]^\top$ denote the global inducing outputs at inducing inputs $\{\tilde{\mathbf{x}}_\ell\}_{\ell=1}^{L_2}$ with covariance $\mathbf{K}_R^{(k)}$ and cross-covariance $\mathbf{K}_{jR}^{(k)} = [C(\mathbf{x}_*^{(j)}, \tilde{\mathbf{x}}_1), \dots, C(\mathbf{x}_*^{(j)}, \tilde{\mathbf{x}}_{L_2})]$. Then the conditional GP prior for each element $w_{kd}^{(j)}$ given $\mathbf{R}_{:kd}$ is

$$p(w_{kd}^{(j)} | \mathbf{R}_{:kd}, \Theta_W) = \mathcal{N}\left(\mathbf{K}_{jR}^{(k)} (\mathbf{K}_R^{(k)})^{-1} \mathbf{R}_{:kd}, s^2 - \mathbf{K}_{jR}^{(k)} (\mathbf{K}_R^{(k)})^{-1} \mathbf{K}_{jR}^{(k)}\right). \quad (4)$$

Integrating out $\mathbf{R}$ under the Gaussian $q(\mathbf{R})$ yields the marginal

$$q(\mathbf{W}_j) = \int p(\mathbf{W}_j | \mathbf{R}, \Theta_W) q(\mathbf{R}) d\mathbf{R} = \mathcal{N}(\mathbf{W}_j | \boldsymbol{\mu}_W^{(j)}, \boldsymbol{\Sigma}_W^{(j)}), \quad (5)$$

whose moments follow from the conditional-Gaussian propagation formulas:

$$\begin{aligned} \boldsymbol{\mu}_W^{(j)}(k, d) &= \mathbf{K}_{jR}^{(k)} (\mathbf{K}_R^{(k)})^{-1} \boldsymbol{\mu}_{kd}, \\ \boldsymbol{\Sigma}_W^{(j)}(k, d) &= s^2 - \mathbf{K}_{jR}^{(k)} (\mathbf{K}_R^{(k)})^{-1} \mathbf{K}_{jR}^{(k)} + \mathbf{K}_{jR}^{(k)} (\mathbf{K}_R^{(k)})^{-1} \boldsymbol{\Sigma}_{kd} (\mathbf{K}_R^{(k)})^{-1} \mathbf{K}_{jR}^{(k)}, \end{aligned} \quad (6)$$

where $\boldsymbol{\mu}_{kd}$ and $\boldsymbol{\Sigma}_{kd}$ are the mean vector, and $\mathbf{R}_{:kd} := (R_{1kd}, \dots, R_{L_2, k, d})^\top \in \mathbb{R}^{L_2}$ denotes the slice of $\mathbf{R}$ along the inducing-point index ℓ for fixed (k, d) . Based on the posterior (1), we have $q(\mathbf{R}_{:kd}) = \mathcal{N}(\boldsymbol{\mu}_{kd}, \boldsymbol{\Sigma}_{kd})$ with $\boldsymbol{\mu}_{kd} = (\mu_{1kd}, \dots, \mu_{L_2, k, d})^\top$ and $\boldsymbol{\Sigma}_{kd} = \text{diag}(\sigma_{1kd}^2, \dots, \sigma_{L_2, k, d}^2)$.

ELBO decomposition. Using Jensen's inequality, the evidence lower bound (ELBO) can be written as

$$\begin{aligned} \mathcal{L} &= \sum_{j=1}^J \underbrace{\left(\mathbb{E}_{q(\mathbf{r}^{(j)})q(\mathbf{W}_j)q(\mathbf{v}^{(j)})} \left[\log p(\mathbf{y}^{(j)} | \mathbf{v}^{(j)}, \mathbf{f}^{(j)}, \Theta^{(j)}) \right] + \mathbb{E}_{q(\mathbf{W}_j)q(\mathbf{v}^{(j)})} \left[\log p(\mathbf{v}^{(j)} | \Theta^{(j)}) - \log q(\mathbf{v}^{(j)}) \right] \right)}_{\text{(I) Likelihood and partition term}} \\ &\quad - \sum_{j=1}^J \underbrace{\text{KL}(q(\mathbf{r}^{(j)}) \| p(\mathbf{r}^{(j)}))}_{\text{(II) Function prior regularization}} - \underbrace{\text{KL}(q(\mathbf{R}) \| p(\mathbf{R} | \Theta_W))}_{\text{(III) Projection prior regularization}}. \end{aligned} \quad (7)$$

The first group (I) corresponds to the expected local data likelihood and latent-indicator partition term, the second group (II) regularizes each region's inducing variable posterior toward its GP prior, and the third group (III) penalizes deviation of the global projection posterior $q(\mathbf{R})$ from its GP prior parameterized by Θ_W .

(I) Likelihood term: details for a fixed region j

The conditional likelihood is

$$\log p(\mathbf{y}^{(j)} \mid \mathbf{f}^{(j)}, \mathbf{v}^{(j)}) = \sum_{i \in \mathcal{D}_n^{(j)}} \left[v_i^{(j)} \log \mathcal{N}(y_i^{(j)} \mid f_i^{(j)}, \sigma_j^2) + (1 - v_i^{(j)}) \log \frac{1}{u_j} \right].$$

GP conditional for $\mathbf{f}^{(j)}$. With local inducing variables $\mathbf{r}^{(j)}$ (standardized outputs) and projection $\mathbf{W}_j$, the conditional prior is

$$p(\mathbf{f}^{(j)} \mid \mathbf{r}^{(j)}, \mathbf{W}_j) = \mathcal{N}(\mathbf{K}_{fr}^{(j)} (\mathbf{K}_r^{(j)})^{-1} \mathbf{r}^{(j)}, a_{m(j)} \mathbf{C}_{nn}^{(j)} - \mathbf{K}_{fr}^{(j)} (\mathbf{K}_r^{(j)})^{-1} \mathbf{K}_{rf}^{(j)}),$$

where $[\mathbf{K}_{fr}^{(j)}]_{i\ell} = a_{m(j)} C(\|\mathbf{W}_j \mathbf{x}_i^{(j)} - \tilde{\mathbf{z}}_\ell^{(j)}\|^2)$ and $\mathbf{C}_{nn}^{(j)}$ is built from projected local inputs $\{\mathbf{W}_j \mathbf{x}_i^{(j)}\}$.

For each i ,

$$\begin{aligned} \mathbb{E}[f_i^{(j)} \mid \mathbf{r}^{(j)}, \mathbf{W}_j] &= \mathbf{K}_{fr}^{(i,j)} (\mathbf{K}_r^{(j)})^{-1} \mathbf{r}^{(j)}, \\ \text{Var}(f_i^{(j)} \mid \mathbf{r}^{(j)}, \mathbf{W}_j) &= a_{m(j)} - \mathbf{K}_{fr}^{(i,j)} (\mathbf{K}_r^{(j)})^{-1} \mathbf{K}_{rf}^{(i,j)}. \end{aligned}$$

Taking expectation over $q(\mathbf{r}^{(j)})$ and $q(\mathbf{W}_j)$ gives

$$\begin{aligned} \mathbb{E}_{q(\mathbf{W}_j)q(\mathbf{v}^{(j)})}[f_i^{(j)}] &= \mathbb{E}_{q(\mathbf{W}_j)}[\mathbf{K}_{fr}^{(i,j)}] (\mathbf{K}_r^{(j)})^{-1} \boldsymbol{\mu}_r^{(j)}, \\ \mathbb{E}_{q(\mathbf{W}_j)q(\mathbf{v}^{(j)})}[(f_i^{(j)})^2] &= \mathbb{E}_{q(\mathbf{W}_j)}[\text{Var}(f_i^{(j)} \mid \mathbf{r}^{(j)}, \mathbf{W}_j)] + \mathbb{E}_{q(\mathbf{W}_j)}[(\mathbf{K}_{fr}^{(i,j)} (\mathbf{K}_r^{(j)})^{-1} \boldsymbol{\mu}_r^{(j)})^2] \\ &\quad + \text{tr}(\mathbb{E}_{q(\mathbf{W}_j)}[\mathbf{K}_{fr}^{(i,j)} (\mathbf{K}_r^{(j)})^{-1} \mathbf{K}_{rf}^{(i,j)}] (\mathbf{K}_r^{(j)})^{-1} \boldsymbol{\Sigma}_r^{(j)}). \end{aligned}$$

Closed forms via kernel expectations. Introduce

$$\begin{aligned} \Psi_1^{(j)} &= \mathbb{E}_{q(\mathbf{W}_j)}[\mathbf{K}_{fr}^{(j)}] \in \mathbb{R}^{n \times L_1}, \\ \Psi_2^{(i,j)} &= \mathbb{E}_{q(\mathbf{W}_j)}[\mathbf{K}_{rf}^{(i,j)} \mathbf{K}_{fr}^{(i,j)}] \in \mathbb{R}^{L_1 \times L_1}. \end{aligned} \tag{8}$$

where we denote $\mathbf{K}_{fr}^{(i,j)} \in \mathbb{R}^{1 \times L_1}$ for the i -th row of $\mathbf{K}_{fr}^{(j)}$, i.e. $\mathbf{K}_{fr}^{(i,j)} \triangleq [\mathbf{K}_{fr}^{(j)}]_{i,:}$, and accordingly $\mathbf{K}_{rf}^{(i,j)} \triangleq (\mathbf{K}_{fr}^{(i,j)})^\top \in \mathbb{R}^{L_1 \times 1}$. Assuming a squared-exponential correlation $C(\|\cdot\|)$

$\|\cdot\|^2) = \exp(-\frac{1}{2}\|\cdot\|^2)$ and a mean-field Gaussian marginal for the (k, d) -th entries of $\mathbf{W}_j$,

$$q(w_{kd}^{(j)}) = \mathcal{N}(\mu_{kd}^{(j)}, (\sigma_{kd}^{(j)})^2), \quad (9)$$

the (i, ℓ) entry of $\Psi_1^{(j)}$ admits

$$[\Psi_1^{(j)}]_{i\ell} = a_{m(j)} \prod_{k=1}^K \frac{1}{\sqrt{1 + (\mathbf{x}_i^{(j)})^\top \Sigma_k^{(j)} \mathbf{x}_i^{(j)}}} \exp\left(-\frac{((\boldsymbol{\mu}_k^{(j)})^\top \mathbf{x}_i^{(j)} - \tilde{z}_{\ell k}^{(j)})^2}{2[1 + (\mathbf{x}_i^{(j)})^\top \Sigma_k^{(j)} \mathbf{x}_i^{(j)}]}\right)$$

where $\boldsymbol{\mu}_k^{(j)} \in \mathbb{R}^D$ is the mean vector of row k of $\mathbf{W}_j$ and $\Sigma_k^{(j)} = \text{diag}((\sigma_{k1}^{(j)})^2, \dots, (\sigma_{kD}^{(j)})^2)$ is its diagonal covariance under $q(\mathbf{W}_j)$. Similarly, for $\Psi_2^{(i,j)}$,

$$[\Psi_2^{(i,j)}]_{\ell\ell'} = a_{m(j)}^2 \exp\left(-\frac{1}{2}\|\tilde{\mathbf{z}}_\ell^{(j)} - \tilde{\mathbf{z}}_{\ell'}^{(j)}\|^2\right) \prod_{k=1}^K \frac{1}{\sqrt{1 + 2(\mathbf{x}_i^{(j)})^\top \Sigma_k^{(j)} \mathbf{x}_i^{(j)}}} \exp\left(-\frac{((\boldsymbol{\mu}_k^{(j)})^\top \mathbf{x}_i^{(j)} - \bar{z}_k)^2}{1 + 2(\mathbf{x}_i^{(j)})^\top \Sigma_k^{(j)} \mathbf{x}_i^{(j)}}\right)$$

with $\bar{\mathbf{z}} = (\tilde{\mathbf{z}}_\ell^{(j)} + \tilde{\mathbf{z}}_{\ell'}^{(j)})/2$ and $\bar{z}_k$ its k th component. See Appendix of (AUEB & Lázaro-Gredilla 2013) for a full derivation.

Convenient scalars. For each (j, i) , define

$$Q_{j,i} := \frac{(y_i^{(j)})^2 - 2y_i^{(j)}\zeta_{j,i} + A_{j,i} + B_{j,i}}{2\sigma_j^2},$$

where $\zeta_{j,i}$ denotes the i th element of $\Psi_1^{(j)}(\mathbf{K}_r^{(j)})^{-1}\boldsymbol{\mu}_r^{(j)}$, $A_{j,i} := a_{m(j)} - \text{tr}((\mathbf{K}_r^{(j)})^{-1}\Psi_2^{(i,j)})$, and $B_{j,i} := \text{tr}((\mathbf{K}_r^{(j)})^{-1}\Psi_2^{(i,j)}(\mathbf{K}_r^{(j)})^{-1}(\boldsymbol{\mu}_r^{(j)}\boldsymbol{\mu}_r^{(j)\top} + \Sigma_r^{(j)}))$. Then the expected conditional log-likelihood contribution equals

$$\mathbb{E}_{q(\mathbf{r}^{(j)})q(\mathbf{W}_j)q(\mathbf{v}^{(j)})}[\log p(\mathbf{y}^{(j)} | \mathbf{v}^{(j)}, \mathbf{f}^{(j)}, \boldsymbol{\Theta}^{(j)})] = \sum_{i \in \mathcal{D}_n^{(j)}} \rho_i^{(j)} \left(-\frac{1}{2} \log(2\pi\sigma_j^2) - Q_{j,i}\right).$$

(II) Partitioning expectations

To calculate the term

$$\sum_{i \in \mathcal{D}_n^{(j)}} \mathbb{E}_{q(\mathbf{W}_j)q(\mathbf{v}^{(j)})} \log \frac{p(v_i^{(j)} | \mathbf{x}_i^{(j)}, \mathbf{W}_j, \boldsymbol{\nu}_j)}{q(v_i^{(j)})}.$$

where $p(v_i^{(j)} = 1 | \mathbf{x}_i^{(j)}, \mathbf{W}_j, \boldsymbol{\nu}_j) = \sigma(\xi_i^{(j)})$, with $\xi_i^{(j)} = \boldsymbol{\nu}_j^\top [1, \mathbf{W}_j \mathbf{x}_i^{(j)}]$, we firstly present the explicit form of the posterior distribution of $\xi_i^{(j)}$.

Under the mean-field Gaussian $q(\mathbf{W}_j)$ in (9), denote

$$\mu_{\xi,i}^{(j)} = \nu_{0,j} + \sum_{k=1}^K \sum_{d=1}^D \nu_{k,j} \mu_{kd}^{(j)} x_{i,d}^{(j)}, \quad (\sigma_{\xi,i}^{(j)})^2 = \sum_{k=1}^K \sum_{d=1}^D (\nu_{k,j} x_{i,d}^{(j)})^2 (\sigma_{kd}^{(j)})^2.$$

Hence $\xi_i^{(j)} \sim \mathcal{N}(\mu_{\xi,i}^{(j)}, (\sigma_{\xi,i}^{(j)})^2)$.

Then the expectations $\mathbb{E}_{\xi_i^{(j)}}[\log \sigma(\xi_i^{(j)})]$ and $\mathbb{E}_{z_i^{(j)}}[\log(1 - \sigma(z_i^{(j)}))]$ are computed by Gaussian–Hermite quadrature (Liu & Pierce 1994):

$$\int_{-\infty}^{\infty} e^{-x^2} f(x) dx \approx \sum_{t=1}^{n_q} w_t f(x_t),$$

where x_t are roots of $H_{n_q}(x)$ and the weights are $w_t = \frac{2^{n_q-1} n_q! \sqrt{\pi}}{n_q^2 [H_{n_q-1}(x_t)]^2}$.

(I)+(II) Summary and optimal $q(v)$

Define, for each (i, j) ,

$$\begin{aligned} S_1^{i,j} &= -\frac{1}{2} \log(2\pi\sigma_j^2) - Q_{j,i} + \mathbb{E}_{q(\mathbf{W}_j)} \log \sigma(\boldsymbol{\nu}_j^\top [1, \mathbf{W}_j \mathbf{x}_i^{(j)}]), \\ S_2^{i,j} &= -\log u_j + \mathbb{E}_{q(\mathbf{W}_j)} \log(1 - \sigma(\boldsymbol{\nu}_j^\top [1, \mathbf{W}_j \mathbf{x}_i^{(j)}])). \end{aligned}$$

Then

$$\begin{aligned} (I) + (II) &= \mathbb{E}_{q(\mathbf{r}^{(j)})q(\mathbf{W}_j)q(\mathbf{v}^{(j)})} \left[\log p(\mathbf{y}^{(j)} \mid \mathbf{v}^{(j)}, \mathbf{f}^{(j)}, \boldsymbol{\Theta}^{(j)}) \right] \\ &\quad + \mathbb{E}_{q(\mathbf{W}_j)q(\mathbf{v}^{(j)})} \left[\log p(\mathbf{v}^{(j)} \mid \boldsymbol{\Theta}^{(j)}) - \log q(\mathbf{v}^{(j)}) \right] - \text{KL}(q(\mathbf{r}^{(j)}) \parallel p(\mathbf{r}^{(j)})) \quad (10) \\ &= \sum_{i \in \mathcal{D}_n^{(j)}} \left[\rho_i^{(j)} S_1^{i,j} + (1 - \rho_i^{(j)}) S_2^{i,j} - \rho_i^{(j)} \log \rho_i^{(j)} - (1 - \rho_i^{(j)}) \log(1 - \rho_i^{(j)}) \right]. \end{aligned}$$

Optimizing (10) w.r.t. $\rho_i^{(j)}$ yields

$$\rho_i^{(j)} = \frac{e^{S_1^{i,j}}}{e^{S_1^{i,j}} + e^{S_2^{i,j}}},$$

and the optimal value of (10)

$$(I) + (II) = \sum_{j=1}^J \sum_{i \in \mathcal{D}_n^{(j)}} \log(e^{S_1^{i,j}} + e^{S_2^{i,j}}).$$

(III) KL divergence for the global inducing variables $\mathbf{R}$

From the prior in the main text,

$$p(\mathbf{R}) = \prod_{k=1}^K \prod_{d=1}^D \mathcal{N}(\mathbf{R}_{:kd} \mid \mathbf{0}, \mathbf{K}_R^{(k)}),$$

$$[\mathbf{K}_R^{(k)}]_{\ell\ell'} = s^2 \exp\left(-\frac{\|\tilde{\mathbf{x}}_\ell - \tilde{\mathbf{x}}_{\ell'}\|^2}{2\ell_{w,k}^2}\right).$$

Our *per-element* mean-field posterior is

$$q(\mathbf{R}) = \prod_{\ell,k,d} \mathcal{N}(R_{\ell,k,d} \mid \mu_{\ell kd}, \sigma_{\ell kd}^2) \equiv \prod_{k,d} \mathcal{N}(\boldsymbol{\mu}_{kd}, \boldsymbol{\Sigma}_{kd}),$$

where $\boldsymbol{\mu}_{kd} = [\mu_{1kd}, \dots, \mu_{L_2 kd}]^\top$ and $\boldsymbol{\Sigma}_{kd} = \text{diag}(\sigma_{1kd}^2, \dots, \sigma_{L_2 kd}^2)$. Hence, for each (k, d) ,

$$\boxed{\text{KL}(q(\mathbf{R}_{:kd}) \parallel p(\mathbf{R}_{:kd})) = \frac{1}{2} \left[\log \frac{|\mathbf{K}_R^{(k)}|}{|\boldsymbol{\Sigma}_{kd}|} - L_2 + \text{tr}((\mathbf{K}_R^{(k)})^{-1} \boldsymbol{\Sigma}_{kd}) + \boldsymbol{\mu}_{kd}^\top (\mathbf{K}_R^{(k)})^{-1} \boldsymbol{\mu}_{kd} \right].}$$

Summing over all (k, d) gives the projection prior penalty $\text{KL}(q(\mathbf{R}) \parallel p(\mathbf{R}))$.

Implicit marginal $q(\mathbf{W}_j)$. Since $q(\mathbf{R})$ is Gaussian and $p(\mathbf{W} \mid \mathbf{R})$ is a linear-Gaussian conditional GP, the induced marginal $q(\mathbf{W})$ is Gaussian. In practice, we only need the first two moments of $q(\mathbf{W}_j)$ (entering $\Psi_1^{(j)}$ and $\Psi_2^{(i,j)}$), which are computed analytically from the conditional GP moments and the diagonal $q(\mathbf{R})$ above; the resulting formulas agree with the row-wise mean/variance parameters $\{\mu_{kd}^{(j)}, (\sigma_{kd}^{(j)})^2\}$ used in (I)–(II).

Putting it together and optimization details

Combining (I)–(III) over $j = 1, \dots, J$ yields the full ELBO $\mathcal{L}$. All expectations of $\log \sigma(\cdot)$ are computed by Gaussian-Hermite quadrature with degree n_q ; all remaining expectations are closed-form under the Gaussian assumptions above. We maximize $\mathcal{L}$ by stochastic gradient ascent with respect to

$$\{\boldsymbol{\mu}_r^{(j)}, \boldsymbol{\Sigma}_r^{(j)}\}_{j=1}^J, \quad \{\mu_{\ell kd}, \sigma_{\ell kd}\}_{\ell,k,d}, \quad \tilde{\mathbf{x}} = (\tilde{\mathbf{x}}_\ell)_{\ell=1}^{L_2}, \quad \text{and } \{\nu_j, u_j, \sigma_j, \mu_{m(j)}, a_{m(j)}\}_{j=1}^J, \quad s, \{\ell_{w,k}\}_{k=1}^K.$$

We enforce positivity of variance/lengthscale parameters by optimizing in the log-domain.

Gradients are obtained by automatic differentiation (e.g., PyTorch).

Appendix B. Theoretical Results and Proof

In this appendix, we provide the detailed theoretical analysis and proofs supporting Section 4.

0.1 Oracle Decomposition of the Prediction Error

To separate the squared prediction risk by the error sources, we first introduce four different Oracle predictors. Fix a test anchor $\mathbf{x}_*$ and its neighborhood index set $\mathcal{D}_n^{(*)}$. Let $r(g(\mathbf{x}))$ denote the unknown ground-truth region label induced in the latent space. Define the *true* in-region subset

$$\mathcal{D}_* := \{i \in \mathcal{D}_n^{(*)} : r(g(\mathbf{x}_i)) = r(g(\mathbf{x}_*))\},$$

and let $\hat{\mathcal{D}}_*$ denote the *fitted* (potentially contaminated) gated subset determined by the learned gate parameters (e.g., ν_j). We introduce the four Oracle predictors:

- $\hat{f}_X^{(W)}$: the *JGP predictor* trained on the learned gated neighborhood $\hat{\mathcal{D}}_*$, using the projected inputs with the fitted W and observed outputs, $\{(W\mathbf{x}_i, y_i)\}_{i \in \hat{\mathcal{D}}_*}$
- $\bar{f}_X^{(W)}$: an *oracle GP predictor* trained on correctly gated observations $\{(W\mathbf{x}_i, y_i)\}_{i \in \mathcal{D}_*}$.
- $\bar{f}_X^{(W_*)}$: an *oracle GP predictor* trained on correctly projected and gated observations, $\{(W_*\mathbf{x}_i, y_i)\}_{i \in \mathcal{D}_*}$.
- $\tilde{f}_X^{(W_*)}$: an *aligned-output GP predictor* trained on the hyperthetical data $\{(W_*\mathbf{x}_i, f(W_*\mathbf{x}_i))\}_{i \in \mathcal{D}_*}$.

The difference between $\hat{f}_X^{(W)}$ and $\bar{f}_X^{(W)}$ isolates the effect of mis-classification, i.e., deviation of the learned gated neighborhood $\hat{\mathcal{D}}_*$ from the true gated neighborhood $\mathcal{D}_*$. In contrast, the difference between $\bar{f}_X^{(W)}$ and $\bar{f}_X^{(W_*)}$ is based on the deviation of the fitted projection $\mathbf{W}$ from the true projection W_* . The difference between $\bar{f}_X^{(W_*)}$ and $\tilde{f}_X^{(W_*)}$, on the other hand, represents the locally linear approximation error of $W_*\mathbf{x}$ to $g(\mathbf{x})$ around the test location. Then the prediction error can be decomposed into four terms as follows:

$$\begin{aligned} \hat{f}_X^{(W)} - f(g(\mathbf{x}_*)) &= \underbrace{(\hat{f}_X^{(W)} - \bar{f}_X^{(W)})}_{C_1} + \underbrace{(\bar{f}_X^{(W)} - \bar{f}_X^{(W_*)})}_{C_2} \\ &\quad + \underbrace{(\bar{f}_X^{(W_*)} - \tilde{f}_X^{(W_*)})}_{C_3} + \underbrace{(\tilde{f}_X^{(W_*)} - f(g(\mathbf{x}_*)))}_{C_4}. \end{aligned} \tag{11}$$

Consequently, Let $E_i := \mathbb{E}[C_i^2]$ for $i = 1, \dots, 4$. By the triangle inequality and Cauchy–Schwarz,

$$\mathcal{R} \lesssim E_1 + E_2 + E_3 + E_4.$$

The decomposition isolates four distinct modeling errors: C_1 is the gating/classification error (mis-gating due to imperfect estimates of ν_j); C_2 is the projection estimation error, quantifying the discrepancy between W and the ideal W_* ; C_3 is the local linearization (geometry) error, corresponding to the residual of approximating $g(\cdot)$ by its local linear map near $\mathbf{x}_*$; and C_4 is the standard GP regression estimation error in the latent space.

Lemma 1 (Gating Error E_1). *Under Assumptions 1–3, there exists a constant $C_6 > 0$ such that the error contribution from mis-gating satisfies:*

$$E_1 \leq C_6(\tau^2 + \tau^{-1}\epsilon_n)\Delta_f^2,$$

where $\tau \in (0, 1)$ is a tuning parameter balancing the fraction of out-of-distribution (OOD) points against the probability of large contamination.

Specifically, τ acts as a threshold for the mis-classification event. By choosing the optimal $\tau \asymp \epsilon_n^{1/3}$, the combined gating rate becomes $O(\Delta_f^2 \epsilon_n^{2/3})$. This indicates that the error from false gating becomes negligible relative to the regression error as soon as the gating classifier attains moderate accuracy.

Lemma 2 (Projection Estimation Error E_2). *Under the inducing-point Gaussian process prior on projection matrices, there exist constants $C_1, C_2, C_3 > 0$ such that:*

$$E_2 \leq C_1 K D L_2^{-1} + C_2 \text{KL}(q(R) \| p(R)) + C_3 \|\mathbb{E}_q \mathbf{W} - \mathbf{W}_*\|_F^2.$$

The term $O(K D L_2^{-1})$ represents the error induced by the Nyström approximation (Williams & Seeger 2000, Gittens & Mahoney 2016), which vanishes as the number of global inducing points L_2 increases. In practice, because the ground-truth projection $\mathbf{W}_*$ is unknown, the KL divergence and the structural mismatch term $\|\mathbb{E}_q \mathbf{W} - \mathbf{W}_*\|_F^2$ cannot be evaluated directly. However, the variational optimization of the evidence lower bound (ELBO) implicitly minimizes these components by concentrating the posterior around the most informative local subspaces. Notably, the influences of the inducing point counts (L_1, L_2)

and neighborhood size n are intertwined; while L_2 directly controls the Nyström error, both parameters affect the expressive capacity of the variational posterior and the resulting gating boundary.

Lemma 3 (Local Linearization Error E_3). *There exist constants $C_4, C_5 > 0$ such that the geometric mismatch error satisfies:*

$$E_3 \leq C_4 \mathbb{E}[\rho_r(\mathbf{x}_*)^4] + C_5 \sigma^2.$$

This term arises from approximating the nonlinear map $g(\cdot)$ with its first-order Taylor expansion. The error vanishes under the *local infill* assumption: as the total training size N increases, the density of observations grows such that the neighborhood radius $\rho_r(\mathbf{x}_*)$ of the n nearest neighbors shrinks to zero. Consequently, the second-order remainder $O(\rho_r(\mathbf{x}_*)^4)$ becomes negligible in sufficiently dense regimes.

Lemma 4 (Oracle GP Regression Error E_4). *The statistical complexity of GP regression in the K -dimensional latent space satisfies $E_4 \leq C_7 \text{GP}_{\text{oracle}}(n, K)$, where for a constant $C_7 > 0$:*

$$\text{GP}_{\text{oracle}}(n, K) \lesssim \begin{cases} B_f^2 \frac{(\log n)^{K+1}}{n}, & \text{squared exponential kernel,} \\ B_f^2 n^{-2\nu_M/(2\nu_M+K)}, & \text{Matérn}(\nu_M) \text{ kernel,} \end{cases}$$

where $\nu_M > 0$ denotes the Matérn smoothness parameter

This term represents the finite-sample regression error of a GP (Van der Vaart & Van Zanten 2009, Seeger 2004) trained on n noise-free samples projected via the ideal linear map $\mathbf{W}_*$. It isolates the statistical complexity of learning the function f in the K -dimensional subspace, matching the minimax optimal rates for intrinsic dimension K .

0.2 Main Risk Bound

We now state the main risk bound for DJGP. The bound clarifies how DJGP balances dimensionality reduction, gating accuracy, and local approximation.

Theorem 1 (Overall Risk Bound for DJGP). *Under Assumptions 1–3,*

$$\begin{aligned} \mathcal{R} \lesssim & KD L_2^{-1} + \text{KL}(q(R) \parallel p(R)) + \|\mathbb{E}_q \mathbf{W} - \mathbf{W}_*\|_F^2 + \mathbb{E}[\rho_r(\mathbf{X})^4] + \sigma^2 \\ & + (\tau^2 + \tau^{-1} \epsilon_n) \Delta_f^2 + \text{GP}_{\text{oracle}}(n, K), \end{aligned} \tag{12}$$

where $\tau \in (0, 1)$ and

$$\text{GP}_{\text{oracle}}(n, K) \lesssim \begin{cases} B_f^2 \frac{(\log n)^{K+1}}{n}, & \text{for the squared exponential kernel,} \\ B_f^2 n^{-2\nu_M/(2\nu_M+K)}, & \text{for the Matérn}(\nu_M) \text{ kernel,} \end{cases}$$

with $\nu_M > 0$ the Matérn smoothness parameter.

Our strategy is to bound the four error components in (11) separately. Specifically, we first control E_3 (local linearization error) in Lemma 3, then E_1 (gating error) in Lemma 1, followed by E_2 in Lemma 2, and finally E_4 in Lemma 4. Combining these bounds yields Theorem 1. We also restate and elaborate on several assumptions used in the main text to make the proofs self-contained.

0.3 Proof of Lemma 3

Let $\delta_i^{(W_*)} := z_i^{(W_*)} - g(x_i) = W_* x_i - g(x_i)$ be the *training-input mismatch* at the ideal projection.

Lemma 5 (Neighborhood projection geometry). *Under Assumption 1, for any $i \in \mathcal{D}_n$ with $\|x_i - x_*\| \leq \rho_r(x_*)$,*

$$\left\| \delta_i^{(W_*)}(x_*) \right\| \leq C_g \rho_r(x_*)^2,$$

for a constant C_g depending on the Hessian bound M_g and the local linearization scheme.

Proof.

$$g(x_i) = g(x_*) + g'(x_*)(x_i - x_*) + O((x_i - x_*)^2) = W_* x_i + O((x_i - x_*)^2),$$

since $g'(x_*) = W_*$, $g(x_*) = W_* x_*$ by $g(x_*) = W_* x_* + b_*$. □

Lemma 6 (Projection-induced label mismatch). *For $y_i = f(g(x_i)) + \varepsilon_i$, expand f at $z_i^{(W_*)}$:*

$$f(g(x_i)) = f(z_i^{(W_*)}) - \nabla f(z_i^{(W_*)})^\top \delta_i^{(W_*)}(X) + R_i, \quad |R_i| \leq \frac{1}{2} M_f \left\| \delta_i^{(W_*)}(X) \right\|^2,$$

where M_f bounds the local Hessian of f (inside a region) and R_i is the residual term.

Proposition 1 (Local Lipschitz continuity of f). *By Assumption 2, there exists a radius $R_z > 0$ and a constant $L_f > 0$ such that*

$$|f(z) - f(z')| \leq L_f \|z - z'\|$$

for all $z, z' \in \mathbb{R}^K$ with $\|z\| \leq R_z$ and $\|z'\| \leq R_z$.

Theorem 2. Assume furthermore that the kernel c_m is bounded on the local domain, i.e. there exists $\kappa > 0$ such that

$$|c_m(u, v)| \leq \kappa \quad \text{for all } u, v \text{ with } \|u\| \leq R_z, \|v\| \leq R_z,$$

and that the neighborhood size n is uniformly bounded, $n(x) \leq k_{\max}$ for all x . Then, for any fixed $(x_*, \mathcal{D}_X)$, we have the conditional bound

$$\mathbb{E}[(\bar{f}_X^{(W^*)} - \tilde{f}_X^{(W^*)})^2 | x_*, \mathcal{D}_X] \leq C_2 \rho_r(x_*)^4 + C_3 \sigma^2, \quad (13)$$

where the constants

$$C_2 = 2 L_f^2 C_g^2 C_\alpha, \quad C_3 = 2 C_\alpha,$$

and $C_\alpha := \kappa^2 k_{\max} \sigma^{-4}$ do not depend on n .

Proof. Fix $(x_*, \mathcal{D}_X)$ and the corresponding neighborhood $\mathcal{D}_*$. Recall that the observations satisfy $y_i = f(g(x_i)) + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ independent across i . Define the projection-induced label error and the observational noise contributions by

$$\varepsilon_i^{\text{proj},*} := f(g(x_i)) - f(z_i^{(W^*)}), \quad \varepsilon_i^{\text{obs}} := \varepsilon_i.$$

Recall that the GP posterior mean at x_* under the aligned inputs $z_i^{(W^*)}$ admits the standard kernel-ridge form $\bar{f}_X^{(W^*)} = k_*^\top (K + \sigma^2 I)^{-1} y$, where $(K)_{ij} = k(z_i^{(W^*)}, z_j^{(W^*)})$ and $(k_*)_i = k(z_*^{(W^*)}, z_i^{(W^*)})$. Define the corresponding weights $\alpha^{(W^*)} := (K + \sigma^2 I)^{-1} k_*$, i.e., $\alpha_i^{(W^*)} = e_i^\top (K + \sigma^2 I)^{-1} k_*$.

Then we can rewrite the local GP predictor at W^* as

$$\bar{f}_X^{(W^*)} = \sum_{i \in N_k(X)} \alpha_i^{(W^*)} (f(z_i^{(W^*)}) + \varepsilon_i^{\text{proj},*} + \varepsilon_i^{\text{obs}}),$$

while the aligned-data predictor is

$$\tilde{f}_X^{(W^*)} = \sum_{i \in N_k(X)} \alpha_i^{(W^*)} f(z_i^{(W^*)}).$$

Hence their difference can be written as

$$\bar{f}_X^{(W^*)} - \tilde{f}_X^{(W^*)} = \sum_{i \in N_k(X)} \alpha_i^{(W^*)} (\varepsilon_i^{\text{proj},*} + \varepsilon_i^{\text{obs}}).$$

We first bound the projection-induced errors $\varepsilon_i^{\text{proj},*}$. By Assumption 2 and Lemma 5, we have

$$|\varepsilon_i^{\text{proj},*}| = |f(g(x_i)) - f(z_i^{(W^*)})| \leq L_f \|g(x_i) - z_i^{(W^*)}\| = L_f \|\delta_i^{(W^*)}(x_*)\| \leq L_f C_g \rho_r(x_*)^2.$$

Therefore,

$$\sup_{i \in N_k(x_*)} |\varepsilon_i^{\text{proj},*}| \leq L_f C_g \rho_r(x_*)^2.$$

Next, we control the squared norm of the weight vector $\alpha^{(W^*)}$. By definition,

$$\alpha^{(W^*)} = (K(W^*) + \sigma^2 I)^{-1} k_*(W^*),$$

and since $K(W^*)$ is positive semi-definite, we have

$$\|(K(W^*) + \sigma^2 I)^{-1}\|_{\text{op}} \leq \frac{1}{\sigma^2}.$$

On the other hand, by the boundedness of the kernel on the local domain,

$$\|k_*(W^*)\|_2^2 = \sum_{i \in N_k(X)} k(z_*^{(W^*)}, z_i^{(W^*)})^2 \leq \kappa^2 k(x_*) \leq \kappa^2 k_{\max}.$$

Combining the two inequalities yields

$$\|\alpha^{(W^*)}\|_2 = \|(K(W^*) + \sigma^2 I)^{-1} k_*(W^*)\|_2 \leq \frac{1}{\sigma^2} \|k_*(W^*)\|_2 \leq \frac{\kappa \sqrt{k_{\max}}}{\sigma^2}.$$

Thus

$$\sum_{i \in N_k(X)} (\alpha_i^{(W^*)})^2 = \|\alpha^{(W^*)}\|_2^2 \leq C_\alpha := \frac{\kappa^2 k_{\max}}{\sigma^4}.$$

We now bound the conditional mean squared error $\Delta_X := \bar{f}_X^{(W^*)} - \tilde{f}_X^{(W^*)}$. Using $(a+b)^2 \leq 2(a^2 + b^2)$ and conditioning on $(x_*, \mathcal{D}_X)$, we obtain

$$\mathbb{E}[\Delta_X^2 | x_*, \mathcal{D}_X] \leq 2 \mathbb{E}\left[\left(\sum_{i \in N_k(X)} \alpha_i^{(W^*)} \varepsilon_i^{\text{proj},*}\right)^2 \middle| X, \mathcal{D}_X\right] + 2 \mathbb{E}\left[\left(\sum_{i \in N_k(X)} \alpha_i^{(W^*)} \varepsilon_i^{\text{obs}}\right)^2 \middle| X, \mathcal{D}_X\right].$$

For the first term, we use the uniform bound on $\varepsilon_i^{\text{proj},*}$:

$$\left|\sum_{i \in N_k(x_*)} \alpha_i^{(W^*)} \varepsilon_i^{\text{proj},*}\right| \leq \sup_{i \in N_k(x_*)} |\varepsilon_i^{\text{proj},*}| \sum_{i \in N_k(x_*)} |\alpha_i^{(W^*)}| \leq L_f C_g \rho_r(x_*)^2 \|\alpha^{(W^*)}\|_2 \sqrt{k(X)},$$

and hence

$$\left(\sum_{i \in N_k(X)} \alpha_i^{(W^*)} \varepsilon_i^{\text{proj},*} \right)^2 \leq L_f^2 C_g^2 \rho_r(x_*)^4 \|\alpha^{(W^*)}\|_2^2 k(X) \leq L_f^2 C_g^2 \rho_r(x_*)^4 C_\alpha k_{\max}.$$

Since this bound is deterministic given $(x_*, \mathcal{D}_X)$, it also bounds the conditional expectation.

Absorbing $k_{\max}$ into the constant yields the first part of (13) with $2L_f^2 C_g^2 C_\alpha$.

For the second term, we use the independence and zero-mean of the observational noises $(\varepsilon_i^{\text{obs}})_i$:

$$\mathbb{E} \left[\left(\sum_{i \in N_k(X)} \alpha_i^{(W^*)} \varepsilon_i^{\text{obs}} \right)^2 \middle| x_*, \mathcal{D}_X \right] = \sum_{i \in N_k(X)} (\alpha_i^{(W^*)})^2 \mathbb{E}[(\varepsilon_i^{\text{obs}})^2] = \sigma^2 \sum_{i \in N_k(X)} (\alpha_i^{(W^*)})^2 \leq \sigma^2 C_\alpha.$$

Multiplying by the outer factor 2 gives the second part of (13) with $2C_\alpha \sigma^2$.

Finally, taking expectation of (13) over $(x_*, \mathcal{D}_X)$ yields

$$E_3 = \mathbb{E}[\mathbb{E}[\Delta_X^2 \mid x_*, \mathcal{D}_X]] \leq 2L_f^2 C_g^2 C_\alpha \mathbb{E}[\rho_r(x_*)^4] + 2C_\alpha \sigma^2,$$

which is of the desired form with $C_2 = 2L_f^2 C_g^2 C_\alpha$ and $C_3 = 2C_\alpha$. $\square$

0.4 Proof of Lemma 1

Lemma 7 (JGP prediction error under small contamination). *Fix a test location x_* and its neighbourhood $\mathcal{D}_n^*$. Let the local JGP predictor be*

$$\hat{f}_X = \sum_{i \in \hat{\mathcal{D}}_*} \alpha_i y_i,$$

where the weights α_i are computed from the kernel matrix and test kernel vector at projection W^* . Assume:

- (Bounded labels) There exists $\Delta_f > 0$ such that $|y_i| \leq \Delta_f$ almost surely for all i in the neighborhood.
- (Uniform weight bound)

$$\sum_{i \in \hat{\mathcal{D}}_*} |\alpha_i| \leq C_\alpha,$$

where C_α does not depend on n .

Denote

- (Gating error indicators) Let $I_i := \mathbf{1}\{\hat{r}(g(x_i)) \neq r(g(x_i))\}$, $M := \{i : I_i = 1\}$, $m := |M|$.
- (Small contamination event) For a fixed $\tau \in (0, 1)$ define

$$\mathcal{C}_\tau := \left\{ \theta_X := \frac{\sum_{i \in M} |\alpha_i|}{\sum_{i \in N_k(X)} |\alpha_i|} \leq \tau \right\}.$$

Thus θ_X is the fraction of JGP weight falling on OOD points.

Let the “oracle” predictor (using only correctly-gated points) be

$$\bar{f}_X^{\text{oracle}} := \sum_{i \in \mathcal{D}_*} \alpha_i y_i.$$

Then the squared prediction error between JGP and Oracle GP satisfies

$$E_1 = \mathbb{E} \left[\left(\hat{f}_X^{(W)} - \bar{f}_X^{(W)} \right)^2 \right] \leq \tau^2 C_\alpha^2 \Delta_f^2 + 4 \Delta_f^2 \mathbb{P}(\mathcal{C}_\tau^c). \quad (14)$$

The first term quantifies the effect of a small fraction τ of OOD points, and the second term controls the rare large contamination case.

Remark 1 (Replacing the bounded-label condition by sub-Gaussian tails). *The assumption $|y_i| \leq \Delta_f$ a.s. is not compatible with Gaussian noise. It suffices to assume that the latent regression function is bounded, $\sup_x |f(g(x))| \leq B_f$, and the noise variables $\{\varepsilon_i\}$ are independent σ -sub-Gaussian. Then, for any $\delta \in (0, 1)$, by a union bound and the sub-Gaussian tail inequality, with probability at least $1 - \delta$,*

$$\max_{i \in N_k(x_*)} |y_i| \leq B_f + \sigma \sqrt{2 \log \frac{2k}{\delta}} =: \Delta_f(\delta).$$

Consequently, every step in the proof of Lemma 7 that uses $|y_i| \leq \Delta_f$ continues to hold on this event by replacing Δ_f with $\Delta_f(\delta)$, yielding a high-probability version of the bound.

Proof. Write the JGP predictor as

$$\hat{f}_X = \sum_{i \notin M} \alpha_i y_i + \sum_{i \in M} \alpha_i y_i = \bar{f}_X^{\text{oracle}} + E_{\text{cont}},$$

where the contamination term $E_{\text{cont}} := \sum_{i \in M} \alpha_i y_i$.

Since $|y_i| \leq \Delta_f$ and $\sum_i |\alpha_i| \leq C_\alpha$, we can bound E_{cont} by

$$|E_{\text{cont}}| \leq \Delta_f \sum_{i \in M} |\alpha_i| = \Delta_f \theta_X \sum_{i \in N_k(X)} |\alpha_i| \leq \Delta_f \theta_X C_\alpha.$$

We now decompose the total prediction error according to $\mathcal{C}_\tau$:

$$\mathbb{E}[(\hat{f}_X - \bar{f}_X)^2] = \mathbb{E}[(\hat{f}_X - \bar{f}_X)^2 \mathbf{1}_{\mathcal{C}_\tau}] + \mathbb{E}[(\hat{f}_X - \bar{f}_X)^2 \mathbf{1}_{\mathcal{C}_\tau^c}].$$

Good event $\mathcal{C}_\tau$. On $\mathcal{C}_\tau$ we have $\theta_X \leq \tau$, so

$$|E_{\text{cont}}| \leq \tau C_\alpha \Delta_f.$$

$$(\hat{f}_X - f(g(x)))^2 = E_{\text{cont}}^2 \leq \tau^2 C_\alpha^2 \Delta_f^2.$$

Bad event $\mathcal{C}_\tau^c$. On this event we only use the trivial bound

$$|\hat{f}_X - \bar{f}_X| \leq |\hat{f}_X| + |\bar{f}_X| \leq \Delta_f + \Delta_f = 2\Delta_f,$$

so

$$(\hat{f}_X - \bar{f}_X)^2 \leq 4\Delta_f^2.$$

Hence

$$\mathbb{E}[(\hat{f}_X - \bar{f}_X)^2 \mathbf{1}_{\mathcal{C}_\tau^c}] \leq 4\Delta_f^2 \mathbb{P}(\mathcal{C}_\tau^c).$$

□

Assumption 1 (Tsybakov margin and plug-in gating (Audibert & Tsybakov 2007, Tsybakov 2004)). ¹ Let $Z = g(X) \in \mathbb{R}^K$ denote the latent representation, and let $r^*(Z) \in \{0, 1\}$ indicate whether the “correct expert” is active ($r^*(Z) = 1$) or not ($r^*(Z) = 0$). Define the regression function

$$\eta(z) := \mathbb{P}(r^*(Z) = 1 \mid Z = z).$$

Assume the following:

1. (Tsybakov margin condition) There exist constants $C_0 > 0$ and $\alpha > 0$ such that

$$\mathbb{P}(0 < |\eta(Z) - \tfrac{1}{2}| \leq t) \leq C_0 t^\alpha \quad \text{for all } t > 0. \quad (15)$$

2. (Plug-in gating rule) The gating classifier is a plug-in rule of the form

$$\hat{r}^*(z) = \mathbf{1}\{\hat{\eta}(z) \geq \tfrac{1}{2}\},$$

where $\hat{\eta}$ is an estimator of η depending on some “gating sample size” n .

¹Assumption 1 is a detailed version of Assumption 3.

3. (Regression estimation error) There exist a sequence $\varepsilon_n \downarrow 0$ and a constant $C_\eta > 0$ such that

$$\mathbb{E}[|\hat{\eta}(Z) - \eta(Z)|^{1+\alpha}] \leq C_\eta \varepsilon_n^{1+\alpha}. \quad (16)$$

Then, by standard plug-in classification theory under Tsybakov noise, there exists a constant $C_T > 0$ (depending only on C_m and α) such that the misclassification probability of the gating rule satisfies

$$\mathbb{P}(\hat{r}^*(Z) \neq r^*(Z)) \leq C_T \mathbb{E}[|\hat{\eta}(Z) - \eta(Z)|^{1+\alpha}] \leq C_T C_\eta \varepsilon_n^{1+\alpha} =: \epsilon_n. \quad (17)$$

Lemma 8 (Probabilistic control of the contamination event). *Under Assumption 1, assume that, conditional on the latent features $\{g(x_i)\}_{i \in N_k(x_*)}$, the variables $(I_i)_{i \in N_k(x_*)}$ are independent and each has*

$$\mathbb{P}(I_i = 1 \mid g(x_i)) \leq \epsilon_n$$

with ϵ_n as in (17). Then, for any x_* and any $\tau \in (0, 1)$,

$$\mathbb{P}(\mathcal{C}_\tau^c) \leq \frac{\epsilon_n}{\tau}. \quad (18)$$

In particular, if $\varepsilon_n \asymp n^{-\beta}$ for some $\beta > 0$ in (16), then

$$\mathbb{P}(\mathcal{C}_\tau^c) \lesssim \frac{1}{\tau} n^{-\beta(1+\alpha)}.$$

Proof. Condition on the latent features $\{g(x_i)\}_{i \in N_k(x_*)}$ where $N_k(x_*)$ is the index set of the $n(x_*)$ nearest neighborhood of x_* , by assumption, the I_i are independent Bernoulli variables with $\mathbb{E}[I_i \mid g(x_i)] \leq \epsilon_n$ and

$$m = \sum_{i \in \mathcal{D}_n^*} I_i.$$

First note that, deterministically,

$$\theta_X = \frac{\sum_{i \in M} |\alpha_i|}{\sum_{i \in N_k(x_*)} |\alpha_i|} \leq \frac{\sum_{i \in M} |\alpha_i|}{\min_{j \in N_k(x_*)} |\alpha_j|} \frac{1}{n(x_*)} \leq \frac{m}{n(x_*)},$$

provided all $\alpha_j \neq 0$; if some $\alpha_j = 0$, the inequality is even easier since those indices do not contribute to the numerator. Hence

$$\{\theta_X > \tau\} \subseteq \left\{ \frac{m}{n(x_*)} > \tau \right\}$$

and therefore

$$\mathbb{P}(\mathcal{C}_\tau^c) = \mathbb{P}(\theta_X > \tau) \leq \mathbb{P}\left(\frac{m}{n(x_*)} > \tau\right).$$

Applying Markov's inequality conditional on the latent features gives

$$\mathbb{P}\left(\frac{m}{n(x_*)} > \tau \mid \{g(x_i)\}\right) \leq \frac{\mathbb{E}[m/n(x_*) \mid \{g(x_i)\}]}{\tau} = \frac{1}{\tau n(x_*)} \sum_{i \in N_k(x_*)} \mathbb{E}[I_i \mid g(x_i)] \leq \frac{\epsilon_n}{\tau}.$$

Taking expectation with respect to $\{g(x_i)\}$ yields

$$\mathbb{P}\left(\frac{m}{n(x_*)} > \tau\right) \leq \frac{\epsilon_n}{\tau},$$

which is (18). The rate statement follows by substituting $\epsilon_n \leq C_T C_\eta \epsilon_n^{1+\alpha}$ from Assumption 1 and the assumed behavior $\epsilon_n \asymp n^{-\beta}$ into the bound. $\square$

Remark 2 (On the difference between oracle weights). *In Lemma 7, the oracle predictor $\bar{f}_X$ is defined with the same weights $\{\alpha_i\}$ as the JGP predictor, so that the difference $\hat{f}_X - \bar{f}_X$ isolates the effect of training on mis-gated labels. One may ask how this compares to the “true” oracle GP predictor*

$$\tilde{f}_X := \sum_{i \in \mathcal{D}_*} \tilde{\alpha}_i y_i,$$

whose weights $\tilde{\alpha}$ are obtained by recomputing the GP posterior using only the correctly-gated neighborhood $\mathcal{D}_$.*

Let K and $\tilde{K}$ be the Gram matrices (and $k_, \tilde{k}_*$ the test kernel vectors) built from $\hat{\mathcal{D}}_*$ and $\mathcal{D}_*$, respectively, and set*

$$\alpha = (K + \sigma^2 I)^{-1} k_*, \quad \tilde{\alpha} = (\tilde{K} + \sigma^2 I)^{-1} \tilde{k}_*.$$

Using the resolvent identity,

$$(K + \sigma^2 I)^{-1} - (\tilde{K} + \sigma^2 I)^{-1} = (K + \sigma^2 I)^{-1} (\tilde{K} - K) (\tilde{K} + \sigma^2 I)^{-1},$$

we can decompose

$$\alpha - \tilde{\alpha} = (K + \sigma^2 I)^{-1} (k_* - \tilde{k}_*) + (K + \sigma^2 I)^{-1} (\tilde{K} - K) (\tilde{K} + \sigma^2 I)^{-1} \tilde{k}_*.$$

If the kernel is bounded, $|c_m(u, v)| \leq \kappa$, the neighborhood size is uniformly bounded by $k_{\max}$, and at most m points are mis-gated, then

$$\|k_* - \tilde{k}_*\|_2 \lesssim \kappa \sqrt{m}, \quad \|\tilde{K} - K\|_{\text{op}} \lesssim \kappa m,$$

while $\|(K + \sigma^2 I)^{-1}\|_{\text{op}}, \|(\tilde{K} + \sigma^2 I)^{-1}\|_{\text{op}} \leq 1/\sigma^2$. It follows that $\|\alpha - \tilde{\alpha}\|_2 \leq Cm$ and hence $\|\alpha - \tilde{\alpha}\|_1 \leq C'm$ for constants depending only on $(\kappa, \sigma^2, k_{\max})$.

Under the bounded-label condition $|y_i| \leq \Delta_f$, the contribution of this weight perturbation to the prediction error satisfies

$$\left| \sum_{i \in \mathcal{D}_*} (\alpha_i - \tilde{\alpha}_i) y_i \right| \leq \Delta_f \|\alpha - \tilde{\alpha}\|_1 \lesssim \Delta_f m.$$

Since $m \leq \theta_X k_{\max}$, this term is of order $O(\theta_X)$ and thus has the same scaling as the contamination term controlled in Lemma 7. Therefore, treating the oracle predictor as using the same weights $\{\alpha_i\}$ is harmless at the level of the θ_X -rates that enter our final risk bound.

Remark 3 (Independence of gating indicators). *In the proof of Lemma 8, we treat the mis-gating indicators I_i as independent. Strictly speaking, if the gating rule $\hat{h}$ (or its parameters) is learned from the same training sample, then $\{I_i\}$ are not independent due to their shared dependence on $\hat{h}$.*

This assumption can be made exact via a standard sample-splitting (or cross-fitting) scheme: estimate $\hat{h}$ on an independent subsample $\mathcal{D}_{\text{gate}}$ and perform the local GP regression analysis on the remaining subsample $\mathcal{D}_{\text{reg}}$. Conditional on $\mathcal{D}_{\text{gate}}$, the gating rule $\hat{h}$ is fixed, and since the covariates in $\mathcal{D}_{\text{reg}}$ are i.i.d., the resulting indicators $\{I_i\}_{i \in \mathcal{D}_{\text{reg}}}$ are i.i.d. as well, so the concentration steps used in the proof apply verbatim.

Alternatively, without sample splitting, the independence requirement may be relaxed by invoking stability/generalization arguments for the gating estimator: although $\{I_i\}$ are dependent, the proof only requires concentration for $\sum_i I_i$, which can be controlled under mild stability conditions, leading to the same order of the bound up to constants (and at most logarithmic factors).

Proposition 2 (Combining JGP error and gating rates). *Combining Lemma 7 with Lemma 8, we obtain*

$$\begin{aligned} E_1 &\leq \tau^2 C_\alpha^2 \Delta_f^2 + 4\Delta_f^2 \mathbb{P}(\mathcal{C}_\tau^c) \\ &\leq \tau^2 C_\alpha^2 \Delta_f^2 + \frac{4\Delta_f^2}{\tau} \epsilon_n, \end{aligned}$$

where $\epsilon_n \leq C_T C_\tau \varepsilon_n^{1+\alpha}$ is determined by the regression estimation error of the gating model under the Tsybakov margin condition.

0.5 Proof of Lemma 2

In this subsection, we will bound the term $E_2 = \mathbb{E} \|\hat{f}^{(W)} - \hat{f}^{(W_*)}\|^2$.

Let $c(\cdot, \cdot) : \mathbb{R}^K \times \mathbb{R}^K \rightarrow \mathbb{R}$ be a positive definite kernel (e.g. squared exponential or Matérn) and consider the standard GP regression model with Gaussian likelihood

$$f \sim \mathcal{GP}(0, c), \quad y_i = f(z_i(W)) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

Given W , the posterior mean of f at the test input x can be written as

$$\hat{f}^{(W)}(x) = k_W(x, X)^\top \alpha_W, \tag{19}$$

where

$$k_W(x, X) := (c(z(W), z_1(W)), \dots, c(z(W), z_n(W)))^\top \in \mathbb{R}^n,$$

$$K_W := [c(z_i(W), z_j(W))]_{i,j=1}^n, \quad \alpha_W := (K_W + \sigma^2 I_n)^{-1} y,$$

and $y = (y_1, \dots, y_n)^\top$.

We assume throughout that the inputs are uniformly bounded.

Assumption 2 (Bounded local domain). *There exists $R_x > 0$ such that $\|x_i\| \leq R_x$ and $\|x\| \leq R_x$ for all data points and test inputs considered.*

We also restrict W to a bounded set; this can be seen as conditioning on a high-probability event under the Gaussian prior/posterior.

Assumption 3 (Bounded projection matrices). *There exists $R_W > 0$ such that $\|W\|_{\text{op}} \leq R_W$ and $\|W^*\|_{\text{op}} \leq R_W$.*

Finally we impose a mild regularity assumption on the kernel.

Assumption 4 (Smooth kernel with bounded first derivatives). *The kernel $c(u, v)$ is continuously differentiable in both arguments and there exists $L_k > 0$ such that*

$$\|\nabla_u c(u, v)\| \leq L_k, \quad \|\nabla_v c(u, v)\| \leq L_k$$

whenever $\|u\| \leq R_z$ and $\|v\| \leq R_z$, where $R_z := R_W R_x$ is an upper bound on $\|z_i(W)\|$ and $\|z(W)\|$ implied by Assumptions 2–3.

For standard kernels such as squared exponential or Matérn, the derivatives are bounded on every compact set, hence Assumption 4 holds automatically on the bounded domain specified above.

We first show that, for a fixed data set and a fixed test input x , the posterior mean $\hat{f}^{(W)}(x)$ is a Lipschitz function of W .

0.5.1 Lipschitz continuity in W

Lemma 9 (Lipschitz continuity in W). *Under Assumptions 2–4, there exists a finite constant $C_{\text{loc}} > 0$, depending only on $(n, R_x, R_W, L_k, \sigma^2, \|y\|)$, such that for all $W, W^* \in \mathbb{R}^{K \times D}$ satisfying Assumption 3 we have*

$$|\hat{f}^{(W)}(x) - \hat{f}^{(W^*)}(x)| \leq C_{\text{loc}} \|W - W^*\|_F.$$

Proof. We consider the function

$$F(W) := \hat{f}^{(W)}(x) = k_W(x, X)^\top \alpha_W,$$

with $k_W(x, X)$ and α_W given in (19). By the chain rule,

$$\nabla_W F(W) = (\nabla_W k_W(x, X))^\top \alpha_W + k_W(x, X)^\top \nabla_W \alpha_W. \quad (20)$$

We first bound the two terms on the right-hand side separately.

Step 1: bound on $\nabla_W k_W(x, X)$. For each $i \in \{1, \dots, n\}$ we have

$$k_W(x, x_i) = c(z(W), z_i(W)),$$

with $z(W) = Wx$ and $z_i(W) = Wx_i$. Using the chain rule,

$$\nabla_W k_W(x, x_i) = \nabla_u c(u, v) \big|_{u=z(W), v=z_i(W)} x^\top + \nabla_v c(u, v) \big|_{u=z(W), v=z_i(W)} x_i^\top.$$

By Assumption 4 and the boundedness of x, x_i we obtain

$$\|\nabla_W k_W(x, x_i)\|_F \leq L_k \|x\| + L_k \|x_i\| \leq 2L_k R_x.$$

Stacking the n components we get

$$\|\nabla_W k_W(x, X)\|_F \leq 2nL_k R_x.$$

Step 2: bound on $\nabla_W K_W$ and $\nabla_W \alpha_W$. The (i, j) entry of K_W is $c(z_i(W), z_j(W))$. A calculation analogous to Step 1 yields

$$\|\nabla_W K_W\|_F \leq 4n^2 L_k R_x.$$

Now recall that

$$\alpha_W = (K_W + \sigma^2 I_n)^{-1} y.$$

Differentiating with respect to W gives

$$\nabla_W \alpha_W = -(K_W + \sigma^2 I_n)^{-1} (\nabla_W K_W) (K_W + \sigma^2 I_n)^{-1} y.$$

Since K_W is positive semidefinite and $\sigma^2 > 0$, all eigenvalues of $K_W + \sigma^2 I_n$ are at least σ^2 , hence

$$\|(K_W + \sigma^2 I_n)^{-1}\|_{\text{op}} \leq \frac{1}{\sigma^2}.$$

Consequently

$$\|\nabla_W \alpha_W\|_F \leq \frac{1}{\sigma^4} \|\nabla_W K_W\|_F \|y\| \leq \frac{4n^2 L_k R_x}{\sigma^4} \|y\|.$$

Step 3: bound on $\nabla_W F(W)$. We also need an upper bound for $\|k_W(x, X)\|$. Using positive definiteness and the usual GP prior bound,

$$|c(z(W), z_i(W))| \leq c(z(W), z(W))^{1/2} c(z_i(W), z_i(W))^{1/2} \leq c(0, 0) =: \sigma_f^2.$$

Therefore $\|k_W(x, X)\| \leq \sqrt{n} \sigma_f^2$. Combining this with (20) and the bounds above gives

$$\begin{aligned} \|\nabla_W F(W)\|_F &\leq \|\nabla_W k_W(x, X)\|_F \|\alpha_W\| + \|k_W(x, X)\| \|\nabla_W \alpha_W\|_F \\ &\leq 2n L_k R_x \|(K_W + \sigma^2 I_n)^{-1}\| \|y\| + \sqrt{n} \sigma_f^2 \frac{4n^2 L_k R_x}{\sigma^4} \|y\| \\ &\leq \left(\frac{2n L_k R_x}{\sigma^2} + \frac{4n^{5/2} \sigma_f^2 L_k R_x}{\sigma^4} \right) \|y\| =: C_{\text{loc}}. \end{aligned}$$

Importantly, C_{loc} is independent of W .

Note on the dependence on n . Although the expression for C_{loc} above grows with n , this is only an artefact of the crude bounds used in the intermediate steps. Under mild infill assumption, the neighborhood of x becomes dense in a fixed-radius ball as $n \rightarrow \infty$, which ensures that all quantities entering the derivative—namely $\|k_W(x, X)\|$, $\|\alpha_W\|$,

$\|\nabla_W k_W(x, X)\|_F$, and $\|\nabla_W \alpha_W\|_F$ —remain uniformly bounded in n . Consequently, C_{loc} can be taken to be a constant independent of n . The key observation is that under infill sampling, the empirical Riemann sums

$$\frac{1}{n} \sum_{i=1}^n k(z(W), z_i(W))^2 \quad \text{and} \quad \frac{1}{n} \sum_{i=1}^n \|\nabla_W k(z(W), z_i(W))\|_F$$

converge to finite integrals over the fixed local domain, and the posterior weights satisfy $\|\alpha_W\| = O(1)$ because $\|(K_W + \sigma^2 I)^{-1}\|$ remains uniformly bounded away from 0. These ingredients together imply that $\|\nabla_W F(W)\|_F \leq C_{\text{loc}}$ with C_{loc} independent of n , as formalized in Theorem 1.

Step 4: apply the mean value theorem. Let $W_t := W^* + t(W - W^*)$ for $t \in [0, 1]$. By the fundamental theorem of calculus,

$$F(W) - F(W^*) = \int_0^1 \frac{d}{dt} F(W_t) dt = \int_0^1 \langle \nabla_W F(W_t), W - W^* \rangle_F dt.$$

Using Cauchy–Schwarz,

$$|F(W) - F(W^*)| \leq \int_0^1 \|\nabla_W F(W_t)\|_F \|W - W^*\|_F dt \leq C_{\text{loc}} \|W - W^*\|_F.$$

This yields the desired Lipschitz bound. $\square$

0.5.2 From Lipschitz continuity to a bound in terms of W

We now integrate the pointwise Lipschitz inequality with respect to the variational distribution $q(W)$.

Proposition 3. *Under the assumptions of Lemma 9, for any fixed W^* we have*

$$\mathbb{E}_{q(W)} |\hat{f}^{(W)}(x) - \hat{f}^{(W^*)}(x)|^2 \leq C_{\text{loc}}^2 \mathbb{E}_{q(W)} \|W - W^*\|_F^2.$$

Proof. By Lemma 9,

$$|\hat{f}^{(W)}(x) - \hat{f}^{(W^*)}(x)|^2 \leq C_{\text{loc}}^2 \|W - W^*\|_F^2$$

for every W . Taking expectations with respect to $q(W)$ yields

$$\mathbb{E}_{q(W)} |\hat{f}^{(W)}(x) - \hat{f}^{(W^*)}(x)|^2 \leq C_{\text{loc}}^2 \mathbb{E}_{q(W)} \|W - W^*\|_F^2,$$

which proves the claim. $\square$

To obtain a complete bound we therefore need to control $\mathbb{E}_{q(W)}\|W - W^*\|_F^2$, which we now will analyze.

0.5.3 Bounding $\mathbb{E}\|W - W^*\|_F^2$

We first state a simple identity that decomposes the mean-square error into a variance term and a squared bias.

Lemma 10 (Matrix-valued variance-bias decomposition). *Let X be a random matrix in $\mathbb{R}^{K \times D}$ with finite second moment and let $A \in \mathbb{R}^{K \times D}$ be deterministic. Then*

$$\mathbb{E}\|X - A\|_F^2 = \text{Tr}(\text{Var}(X)) + \|\mathbb{E}X - A\|_F^2.$$

Proof. Write $m := \mathbb{E}X$ and note that $X - A = (X - m) + (m - A)$. Then

$$\|X - A\|_F^2 = \|X - m\|_F^2 + 2\langle X - m, m - A \rangle_F + \|m - A\|_F^2.$$

Taking expectations and using $\mathbb{E}(X - m) = 0$ gives

$$\mathbb{E}\|X - A\|_F^2 = \mathbb{E}\|X - m\|_F^2 + \|m - A\|_F^2.$$

The first term equals the trace of the covariance:

$$\mathbb{E}\|X - m\|_F^2 = \mathbb{E} \text{Tr}((X - m)(X - m)^\top) = \text{Tr}(\text{Var}(X)).$$

This proves the identity. □

Applying Lemma 10 with $X = W$ and $A = W^*$ we obtain

$$\mathbb{E}_{q(W)}\|W - W^*\|_F^2 = \text{Tr}(\text{Var}_q(W)) + \|\mathbb{E}_q W - W^*\|_F^2. \quad (21)$$

We next bound these two terms under the inducing-point GP parameterisation of $q(W)$.

We assume a Gaussian prior and inducing-point representation for the projection process:

- Let $R \in \mathbb{R}^M$ denote the stacked inducing variables (for all latent dimensions and input coordinates).
- The prior joint distribution (W, R) is Gaussian.

- Conditional on R , W is Gaussian with linear mean:

$$W \mid R \sim \mathcal{N}(MR, \Sigma_0),$$

where M is a fixed matrix and Σ_0 does not depend on R . In vector form, with $w = \text{vec}(W)$ and $r = \text{vec}(R)$, this can be written as

$$w \mid r \sim \mathcal{N}(Ar, \Sigma_0)$$

for some matrix A .

- The variational distribution over the inducing variables is Gaussian:

$$q(R) = \mathcal{N}(\mu_R, \Sigma_R).$$

The variational marginal of W is then

$$q(W) = \int p(W \mid R) q(R) \mathrm{d}R.$$

Lemma 11 (Variance under the inducing-point variational family). *Under the assumptions above, the covariance of $w = \text{vec}(W)$ under q satisfies*

$$\text{Var}_q(w) = \Sigma_0 + A\Sigma_RA^\top.$$

Proof. Let $\mathbb{E}_q$ denote expectation with respect to $q(W, R)$. The law of total variance gives

$$\text{Var}_q(w) = \mathbb{E}_q[\text{Var}(w \mid R)] + \text{Var}_q(\mathbb{E}[w \mid R]).$$

By construction, $\text{Var}(w \mid R) = \Sigma_0$ does not depend on R , hence

$$\mathbb{E}_q[\text{Var}(w \mid R)] = \Sigma_0.$$

Furthermore, $\mathbb{E}[w \mid R] = Ar$, so that

$$\text{Var}_q(\mathbb{E}[w \mid R]) = \text{Var}_q(Ar) = A\Sigma_RA^\top,$$

because $r \sim q(R) = \mathcal{N}(\mu_R, \Sigma_R)$. Combining these two identities yields the statement. $\square$

Taking traces in Lemma 11 we obtain

$$\text{Tr}(\text{Var}_q(W)) = \text{Tr}(\Sigma_0) + \text{Tr}(A\Sigma_RA^\top). \quad (22)$$

Assumption 5 (Per-location Nyström conditional variance bound). *For each (k, d) and each region index $j \in \{1, \dots, J\}$, let*

$$\sigma_{0,kd}^{(j)} := \text{Var}([W_j]_{k,d} \mid \mathbf{R}_{:kd}) = \text{Var}(\omega_{k,d}(x_*^{(j)}) \mid \mathbf{R}_{:kd}).$$

We assume that, for the chosen Nyström-type construction of the inducing locations $\{\tilde{x}^{(\ell)}\}_{\ell=1}^{L_2}$ (e.g. leverage-score sampling or kernel k -means), there exists a finite constant $C_{\text{Ny}} > 0$, independent of J, K, D, L_2 , such that (Williams & Seeger 2000, Gittens & Mahoney 2016)

$$\sigma_{0,kd}^{(j)} \leq C_{\text{Ny}} T(L_2), \quad \forall j \in \{1, \dots, J\}, \forall k \in \{1, \dots, K\}, \forall d \in \{1, \dots, D\}. \quad (23)$$

In words: for each scalar projection GP (k, d) and each region j , the Nyström conditional variance at the anchor location $x_^{(j)}$ is bounded by a constant multiple of the Mercer spectral tail $T(L_2)$, independently of J .*

Theorem 3 (Per-region Nyström trace bound for the DJGP projection prior). *Suppose Assumptions 5 hold. Then for any fixed region $j \in \{1, \dots, J\}$, the conditional covariance $\Sigma_0^{(j)} = \text{Cov}(w^{(j)} \mid R)$ of the vectorised projection matrix W_j satisfies*

$$\text{Tr}(\Sigma_0^{(j)}) \leq C_{\text{Ny}} K D T(L_2), \quad (24)$$

where $T(L_2) = \sum_{m>L_2} \lambda_m$ is the Mercer spectral tail of the kernel k .

Proof. By construction, the scalar processes $\{\omega_{k,d}\}_{k,d}$ are mutually independent, and $w^{(j)}$ is formed by stacking the KD scalar entries $[W_j]_{k,d}$. Therefore the conditional covariance $\Sigma_0^{(j)}$ is diagonal (or block-diagonal with 1×1 blocks) in the coordinates indexed by (k, d) , and its trace is given by

$$\text{Tr}(\Sigma_0^{(j)}) = \sum_{k=1}^K \sum_{d=1}^D \text{Var}([W_j]_{k,d} \mid R) = \sum_{k=1}^K \sum_{d=1}^D \sigma_{0,kd}^{(j)}.$$

Applying the per-location Nyström bound (23) to each term in the sum yields

$$\text{Tr}(\Sigma_0^{(j)}) \leq \sum_{k=1}^K \sum_{d=1}^D C_{\text{Ny}} T(L_2) = C_{\text{Ny}} K D T(L_2),$$

which is exactly (24). □

Corollary 1 (Per-region scaling for squared exponential and Matérn kernels). *Under the assumptions of Theorem 3, suppose moreover that the Mercer eigenvalues $(\lambda_m)_{m \geq 1}$ of k satisfy one of the following standard decay conditions:*

(i) **Squared exponential kernel.** There exist constants $C_{\text{SE}}, c_{\text{SE}} > 0$ such that

$$\lambda_m \leq C_{\text{SE}} \exp(-c_{\text{SE}} m^{1/D}), \quad m \geq 1.$$

Then the spectral tail obeys

$$T(L_2) = \sum_{m > L_2} \lambda_m \leq C'_{\text{SE}} \exp(-c'_{\text{SE}} L_2^{1/D})$$

for suitable constants $C'_{\text{SE}}, c'_{\text{SE}} > 0$, and therefore for any region j

$$\text{Tr}(\Sigma_0^{(j)}) \leq C_{\text{Ny}} C'_{\text{SE}} K D \exp(-c'_{\text{SE}} L_2^{1/D}).$$

(ii) **Matérn kernel with smoothness $\nu > 0$.** There exists a constant $C_{\text{M}} > 0$ such that

$$\lambda_m \leq C_{\text{M}} m^{-(2\nu_{\text{M}}+D)/D}, \quad m \geq 1.$$

Then, since $\sum_{m > L_2} m^{-(2\nu_{\text{M}}+D)/D} \asymp L_2^{-2\nu_{\text{M}}/D}$ for $\nu_{\text{M}} > 0$, there exists $C'_{\text{M}} > 0$ with

$$T(L_2) = \sum_{m > L_2} \lambda_m \leq C'_{\text{M}} L_2^{-2\nu_{\text{M}}/D},$$

and hence for any region j

$$\text{Tr}(\Sigma_0^{(j)}) \leq C_{\text{Ny}} C'_{\text{M}} K D L_2^{-2\nu_{\text{M}}/D}.$$

We now control the second term in (22).

Lemma 12. Let $\|\cdot\|_{\text{op}}$ denote the operator norm. Then

$$\text{Tr}(A \Sigma_R A^\top) \leq \|A\|_{\text{op}}^2 \text{Tr}(\Sigma_R).$$

Proof. Note that

$$\text{Tr}(A \Sigma_R A^\top) = \text{Tr}(\Sigma_R^{1/2} A^\top A \Sigma_R^{1/2}) \leq \|A^\top A\|_{\text{op}} \text{Tr}(\Sigma_R) = \|A\|_{\text{op}}^2 \text{Tr}(\Sigma_R),$$

where we used the fact that $\text{Tr}(BC) \leq \|B\|_{\text{op}} \text{Tr}(C)$ for positive semidefinite B, C . $\square$

To further bound $\text{Tr}(\Sigma_R)$ we use the explicit form of the Kullback–Leibler divergence between Gaussians.

Let the prior over R be $p(R) = \mathcal{N}(0, C_R)$ with C_R positive definite. Then the KL divergence between $q(R) = \mathcal{N}(\mu_R, \Sigma_R)$ and $p(R)$ is

$$\text{KL}(q(R) \parallel p(R)) = \frac{1}{2} \left(\text{Tr}(C_R^{-1} \Sigma_R) + \mu_R^\top C_R^{-1} \mu_R - \log \det(C_R^{-1} \Sigma_R) - d_R \right),$$

where d_R is the dimension of R . Since $C_R^{-1} \succeq \lambda_{\min}(C_R^{-1})I$,

$$\text{Tr}(C_R^{-1} \Sigma_R) \geq \lambda_{\min}(C_R^{-1}) \text{Tr}(\Sigma_R).$$

Neglecting the non-negative terms $\mu_R^\top C_R^{-1} \mu_R$ and $-\log \det(C_R^{-1} \Sigma_R) - d_R$, we obtain the inequality

$$\text{KL}(q(R) \parallel p(R)) \geq \frac{1}{2} \lambda_{\min}(C_R^{-1}) \text{Tr}(\Sigma_R),$$

that is,

$$\text{Tr}(\Sigma_R) \leq \frac{2}{\lambda_{\min}(C_R^{-1})} \text{KL}(q(R) \parallel p(R)) = 2\lambda_{\max}(C_R) \text{KL}(q(R) \parallel p(R)). \quad (25)$$

Combining Lemma 12 and (25) yields

$$\text{Tr}(A \Sigma_R A^\top) \leq 2\|A\|_{\text{op}}^2 \lambda_{\max}(C_R) \text{KL}(q(R) \parallel p(R)) =: c_R \text{KL}(q(R) \parallel p(R)), \quad (26)$$

where $c_R := 2\|A\|_{\text{op}}^2 \lambda_{\max}(C_R)$ is a finite constant depending only on the prior and the inducing-point geometry.

Combining (21), (22), we obtain the following result.

Theorem 4 (Bound on $\mathbb{E}\|W - W^*\|_F^2$). *Under the inducing-point parameterisation above, the mean-square error of W admits the upper bound*

$$\mathbb{E}_{q(W)}\|W - W^*\|_F^2 \leq c_W K D L_2^{-1} + c_R \text{KL}(q(R) \parallel p(R)) + \|\mathbb{E}_q W - W^*\|_F^2,$$

where c_W and c_R are finite constants defined in (26). In particular, if $W^* = \mathbb{E}[W \mid y]$ is the exact posterior mean and $q(R)$ is chosen such that $\text{KL}(q(R) \parallel p(R \mid y)) \rightarrow 0$ as $L_2 \rightarrow \infty$, then

$$\mathbb{E}_{q(W)}\|W - W^*\|_F^2 = O(K D L_2^{-1}) \quad \text{as } L_2 \rightarrow \infty.$$

Proof. Equation (21) and (22) give

$$\mathbb{E}_{q(W)}\|W - W^*\|_F^2 = \text{Tr}(\Sigma_0) + \text{Tr}(A \Sigma_R A^\top) + \|\mathbb{E}_q W - W^*\|_F^2.$$

Theorem 3 yields $\text{Tr}(\Sigma_0) \leq c_W K D L_2^{-1}$, and (26) yields $\text{Tr}(A \Sigma_R A^\top) \leq c_R \text{KL}(q(R) \| p(R))$. Substituting these inequalities proves the first bound.

If in addition $W^* = \mathbb{E}[W \mid y]$ and $q(R)$ is chosen so that $\text{KL}(q(R) \| p(R \mid y)) \rightarrow 0$ and $\|\mathbb{E}_q W - W^*\|_F^2 \rightarrow 0$, the Nyström term $c_W K D L_2^{-1}$ then dominates the asymptotic behavior, yielding the stated $O(K D L_2^{-1})$ rate. $\square$

0.6 Proof of Lemma 4

Lemma 13 (Oracle local GP rate). *Let Z be a random latent input in a given region m , drawn from the design distribution on $\mathcal{Z}_m$. Then there exist constants $C_{\text{SE}}, C_{\text{Mat}} > 0$, independent of n and K , such that the following bounds hold.*

1. **Squared exponential kernel.** *If c_m is a squared exponential (RBF) kernel on $\mathcal{Z}_m$, then for all n large enough,*

$$\mathbb{E}[(\tilde{f}_m^{\text{GP}}(Z) - f_m(Z))^2] \leq C_{\text{SE}} B_f^2 \frac{(\log n)^{K+1}}{n}. \quad (27)$$

2. **Matérn kernel.** *If k_j is a Matérn kernel with smoothness parameter $\nu > 0$ on $\mathcal{Z}_j$, then for all n large enough,*

$$\mathbb{E}[(\tilde{f}_m^{\text{GP}}(Z) - f_m(Z))^2] \leq C_{\text{Mat}} B_f^2 n^{-\frac{2\nu_M}{2\nu_M + K}}. \quad (28)$$

Proof. The bounds (27)–(28) are standard GP regression rates on bounded domains under RKHS assumptions; they can be derived from posterior contraction or kernel ridge regression results for squared exponential and Matérn kernels (Van der Vaart & Van Zanten 2009, Seeger 2004), respectively. $\square$

0.7 Proof of Theorem 1

Putting everything together, we get the overall risk bound theorem.

Theorem 5 (Overall risk bound for DJGP). *Let*

$$R := \mathbb{E}[(\hat{f}_X^{(\mathbf{W})} - f(g(\mathbf{x}_*)))^2]$$

denote the prediction risk of DJGP. Under Assumptions 1–3 and the decomposition (11), the four expected terms satisfy

$$\begin{aligned}
E_1 &\leq C_6(\tau^2 + \tau^{-1}\epsilon_n)\Delta_f^2, \\
E_2 &\leq C_1KD L_2^{-1} + C_2\text{KL}(q(R)\|p(R)) + C_3\|\mathbb{E}_q W - W_*\|_F^2, \\
E_3 &\leq C_4\mathbb{E}[\rho_r(x_*)^4] + C_5\sigma^2, \\
E_4 &\leq C_7 \text{GP}_{\text{oracle}}(n, K),
\end{aligned} \tag{29}$$

where n is the neighborhood size and

$$\text{GP}_{\text{oracle}}(n, K) \lesssim \begin{cases} B_f^2 \frac{(\log n)^{K+1}}{n}, & \text{squared exponential kernel,} \\ B_f^2 n^{-2\nu_M/(2\nu_M+K)}, & \text{Matérn}(\nu_M) \text{ kernel.} \end{cases}$$

Thus,

$$\begin{aligned}
R &\leq C_1KD L_2^{-1} + C_2\text{KL}(q(R)\|p(R)) + C_3\|\mathbb{E}_q W - W_*\|_F^2 + C_4\mathbb{E}[\rho_r(x_*)^4] + C_5\sigma^2 \\
&\quad + C_6(\tau^2 + \tau^{-1}\epsilon_n)\Delta_f^2 + C_7 \text{GP}_{\text{oracle}}(n, K).
\end{aligned} \tag{30}$$

Choosing $\eta \asymp \epsilon_n^{1/3}$ yields the combined gating rate

$$(\eta^2 + \eta^{-1}\epsilon_n)\Delta_f^2 \lesssim \Delta_f^2 \epsilon_n^{2/3}.$$

Under the Tsybakov margin condition and a regression estimator satisfying $\epsilon_n \asymp n^{-\beta(1+\alpha)}$, this becomes

$$\Delta_f^2 n^{-\frac{2}{3}\beta(1+\alpha)}.$$

Appendix C. Dimension Expansion Techniques

To investigate how different latent-to-observed mappings g affect the performance of various surrogate models, we design four distinct strategies for modeling the transformation from latent space to the observed input space:

- **Random Projection (RP):** To lift the latent representation of dimension K to a higher-dimensional space D , we generate a full-rank random projection matrix $\mathbf{W} \in \mathbb{R}^{D \times K}$ with entries independently drawn from a standard Gaussian distribution. The resulting transformation is given by $\mathbf{x} = g^{-1}(\mathbf{z}) = \mathbf{W}\mathbf{z} \in \mathbb{R}^D$.

- **Random Fourier Features (RF)**(Rahimi & Recht 2007, Li et al. 2021): RF offers an efficient way to approximate shift-invariant kernels by mapping inputs into a randomized feature space. For RBF kernels, the mapping $g : \mathbb{R}^K \rightarrow \mathbb{R}^{2D}$ is defined as $g^{-1}(\mathbf{z}) := \frac{1}{\sqrt{D}} [\cos \langle \omega_1, \mathbf{z} \rangle, \sin \langle \omega_1, \mathbf{z} \rangle, \dots, \cos \langle \omega_D, \mathbf{z} \rangle, \sin \langle \omega_D, \mathbf{z} \rangle]^T$, where $\omega_i \sim \mathcal{N}(0, \sigma^{-2}I)$. This yields an unbiased approximation to the RBF kernel: $c(\mathbf{z}_i, \mathbf{z}_j) = \exp\left(-\frac{\|\mathbf{z}_i - \mathbf{z}_j\|^2}{2\sigma^2}\right)$. More generally, RF approximates any positive definite shift-invariant kernel using its Fourier transform. In our experiments, we use the following simplified form $g^{-1}(\mathbf{z}) = \sqrt{\frac{2}{D}} \cos(\Omega \mathbf{z} + b)$, where $\Omega \in \mathbb{R}^{D \times K}$ is sampled from $\mathcal{N}(0, I_K)$, and $b \in \mathbb{R}^D$ is drawn uniformly from $[0, 2\pi]$.
- **Polynomial Expansion (PE)**: We enrich the latent representation by including all monomials of the original features up to degree three. This introduces smooth nonlinear interactions while maintaining a controlled feature dimensionality through truncation. Specifically, we define the expanded feature vector as

$$\mathbf{x} = \text{trunc}\left([z_1, \dots, z_K, z_1^2, z_1 z_2, \dots, z_K^2, z_1^3, z_1^2 z_2, \dots, z_K^3]\right),$$

where $\mathbf{z} \in \mathbb{R}^K$ is the original latent vector, and *trunc* denotes selecting the first D components of the full polynomial basis in a fixed order (e.g., lexicographic).

- **Autoencoder (AE)**(Wang et al. 2016, Bank et al. 2023): We employ an over-complete autoencoder to construct synthetic high-dimensional data from the low-dimensional latent variables $\mathbf{z}$. Unlike the typical use of autoencoders for dimensionality reduction, our architecture performs *dimension expansion*, mapping from K to D dimensions with $D > K$. Specifically, the encoder $f_{\text{enc}} : \mathbb{R}^K \rightarrow \mathbb{R}^D$ defines the transformation from latent to observed space, and the decoder $f_{\text{dec}} : \mathbb{R}^D \rightarrow \mathbb{R}^K$ reconstructs back to the latent domain.

The autoencoder is trained directly by minimizing the reconstruction loss $\|\mathbf{z}_i - f_{\text{dec}}(f_{\text{enc}}(\mathbf{z}_i))\|_2^2$, where $\{\mathbf{z}_i\}_{i=1}^N$ represents the latent variable values generated by the simulation process described in Section 4.1.1. After training, only the encoder is retained to generate the observed inputs $\mathbf{x}_i = f_{\text{enc}}(\mathbf{z}_i)$ that serve as the training data for DJGP. Hence, DJGP receives only $\mathbf{x}_i$ (without access to $\mathbf{z}_i$) as input, making this setup a controlled dimension-expansion testbed for evaluating its ability to recover low-dimensional latent structure from high-dimensional data. Note that, unlike

a conventional autoencoder used for dimensionality reduction, our *encoder* network acts as a generator that expands the low-dimensional latent variables $\mathbf{z} \in \mathbb{R}^K$ into high-dimensional observations $\mathbf{x} \in \mathbb{R}^D$ ($D > K$), while the *decoder* reconstructs back to the latent domain. The encoder consists of a linear layer with 64 units, followed by BatchNorm and LeakyReLU, and a final linear layer projecting to D . The decoder mirrors this structure in reverse ($D \rightarrow 64 \rightarrow K$) and omits the final activation. Batch normalization helps mitigate feature sparsity, while LeakyReLU prevents neuron “death.” The model is trained for 100 epochs.

Appendix D. Computation Time

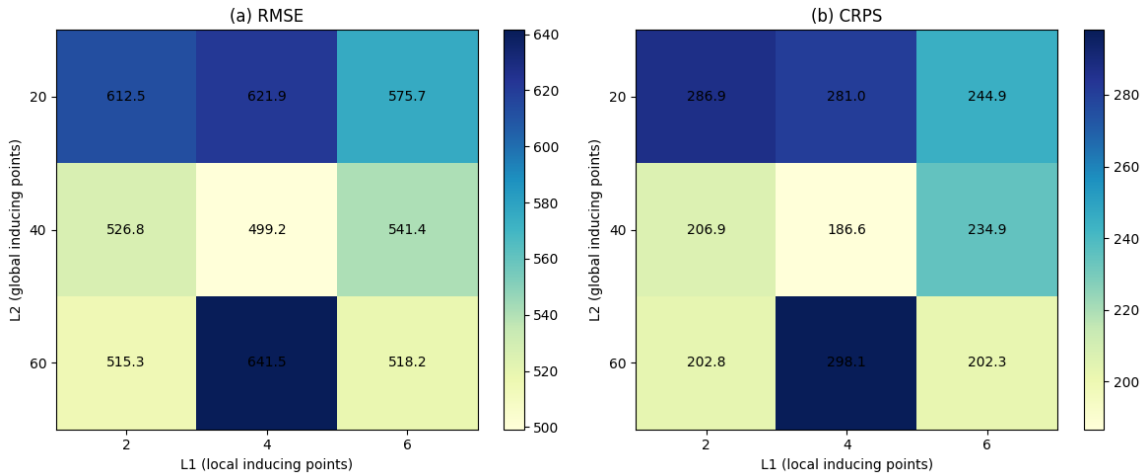


Figure 1: Effect of local and global inducing point counts (L_1, L_2) on (a) RMSE and (b) CRPS. The configuration (4, 40) achieves the best balance between predictive accuracy and uncertainty estimation.

Table 1 reports the average runtime for the real-dataset experiments. JGP-SIR and JGP-PCA benefit from dimensionality reduction, running much faster than JGP. DJGP incurs moderate overhead due to Monte Carlo sampling of projections and repeated local GP inference. Nonetheless, its runtime remains comparable to or lower than DGP, particularly on the large Appliances dataset. DGP exhibits the highest computational cost, stemming from its global variational updates over the entire dataset.

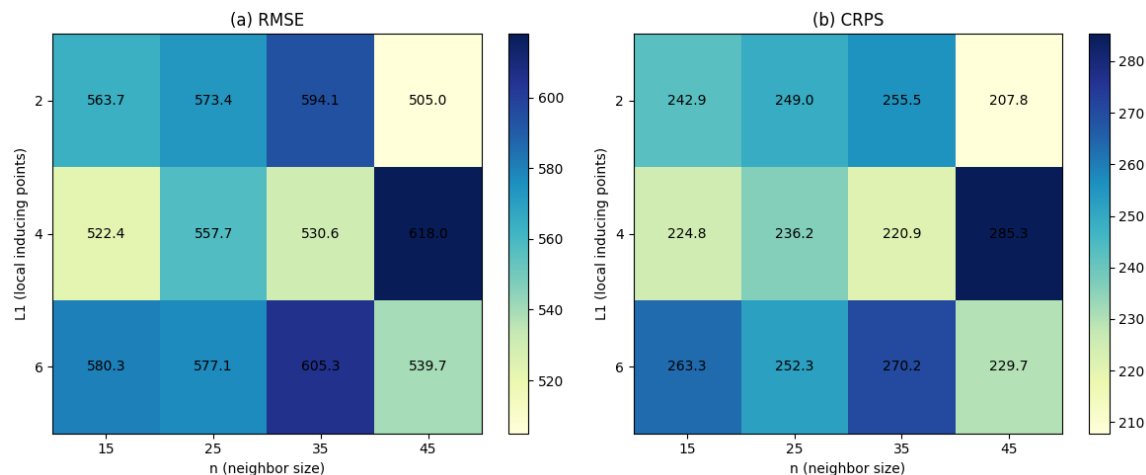


Figure 2: Joint effect of neighborhood size n (horizontal axis) and number of local inducing points L_1 (vertical axis) on (a) RMSE and (b) CRPS.

Dataset	DGP	JGP	JGP-SIR	JGP-PCA	JGP-AE	DJGP _{tr}	DJGP _{inf}	DJGP _{tot}
Wine Quality	179.2	173.1	31.3	32.5	160.9	104.9	97.5	202.4
Parkinson’s Telemonitoring	279.4	384.3	49.7	49.9	206.5	244.0	149.7	393.7
Appliances Energy Prediction	968.2	632.9	52.7	55.7	268.0	199.3	167.1	366.4

Table 1: Average runtime (in seconds) for each method. DJGP_{tr}, DJGP_{inf}, and DJGP_{tot} denote training time, inference time, and total runtime, respectively.

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